

Christine Nattrass
christine.nattrass@utk.edu
(865) 974-9020

- Current Position** • Assistant professor in relativistic heavy ion physics at the University of Tennessee at Knoxville
- Education** • Yale University Physics M.Sc. 2005, M.Phil. 2006, Ph.D. December 2009
• Colorado State University B.S. 2003 cum laude; Physics, Biochemistry, and Physical Science majors; Chemistry, Math, and German minors
- Thesis** • “System, energy, and flavor dependence of jets through di-hadron correlations in heavy ion collisions.” Defended July 20, 2009 Advisor: John Harris.
- External Funding** • Collaborative research: Conferences for Undergraduate Women in Physics for \$8,000 from the NSF November 2011 – September 2012 (co-PI Adriana Moreo)
• Subcontract of Conferences for Undergraduate Women in Physics grant from the DOE November 2011 – November 2012 (PI Sarah Demers, Yale) for \$6,200
- Collaboration Memberships** • The ALICE collaboration – August 2009 – present
• The PHENIX collaboration – August 2009 – present
• The ALICE EMCal collaboration – July 2008 - present
• The STAR collaboration – June 2003 – August 2009
- Publications** • h-index: 23
• Paper on transverse energy in ALICE in preparation. Leading analysis.
• “Near-side azimuthal and pseudo-rapidity correlations of strange baryons and mesons in d+Au, Cu+Cu and Au+Au collisions at $\sqrt{s_{NN}} = 200$ GeV” (STAR collaboration) in preparation for Phys. Rev. C. Did analysis and writing and wrote analysis note with Jana Bielcikova.
• “D-meson production in p-Pb collisions at $\sqrt{s_{NN}} = 5.02$ TeV” (ALICE Collaboration) to be submitted to PRL. On Internal Review Committee.
• “ $K^*(892)^0$ and $\Phi(1020)$ production in Pb-Pb collisions at $\sqrt{s_{NN}} = 2.76$ TeV” (ALICE Collaboration) Phys. Rev. C **91**, 024609 (2015). On Internal Review Committee.
• “Heavy-flavor electron-muon correlations in p+p and d+Au collisions at $\sqrt{s_{NN}} = 200$ GeV,” (PHENIX Collaboration) Phys. Rev. C. **89**, 034915 (2014) in preparation for PRC. Paper Preparation Committee.
• “System size and energy dependence of near-side di-hadron correlations” (STAR collaboration) Phys. Rev. C **85** (2012) 014903. Did the majority of the analysis and writing and wrote the majority of the internal analysis note.
• “Particle yield modification in jet-like azimuthal di-hadron correlations in Pb-Pb collisions at $\sqrt{s_{NN}} = 2.76$ TeV” (ALICE collaboration) accepted by Phys. Rev. Lett (January 2012). On internal review committee.
• “Measurements of Bottom Quark Production from electron hadron Angular Correlations in pp Collisions at $\sqrt{s} = 200$ GeV,” (STAR Collaboration) Phys. Rev. Lett. **105**, 202301 (2010). On internal Godparent committee.
• “An Experimental Exploration of the QCD Phase Diagram: The Search for the Critical Point and the Onset of De-confinement,” (STAR Collaboration) arXiv:1007.2613. Performed calculations for high-pt triggered di-hadron correlations.
• “System size dependence of associated yields in hadron-triggered jets,” (STAR Collaboration) Phys. Lett. B **683** (2010) 123. Helped with Cu+Cu data analysis.
• “ALICE EMCAL Physics Performance Report”, (ALICE Collaboration) arXiv:1008.0413v1. Helped with analysis for heavy flavor capabilities.
• “ALICE electromagnetic calorimeter technical design report” (ALICE Collaboration) CERN-LHCC-2008-014, CERN-ALICE-TDR-014, (2008). Part of ALICE EMCal Collaboration.
• “Performance of prototypes for the ALICE electromagnetic calorimeter,” Nuclear Inst. and Methods in Physics Research A **615** (2010) 6-13. Part of ALICE EMCal Collaboration.
• “Elastic properties of $\text{LaSn}_x\text{Ni}_{5-x}$ compounds,” (O. Yeheskel, C. E. Nattrass, R. G. Leisure, I. Jacob, R. C. Bowman), J Appl Phys **95** (3): 1089-1095 (2004).
• “Temperature dependence of the elastic moduli of polycrystalline $\text{LaAl}_x\text{Ni}_{5-x}$ and $\text{LaSn}_x\text{Ni}_{5-x}$,” (J.E.

Atteberry, D.S. Agosta, C.E. Nattrass, R. G. Leisure, I. Jacob, R. C. Bowman) Journal of Alloys and Compounds, Volume 376, Issues 1-2, 139. (2004).

• [Full publication list available upon request](#)

- Conference Proceedings**
- “Measurements of jets in ALICE,” C. Nattrass (ALICE), *J. Phys.: Conf. Ser.* **589** 011015, Proceedings for 9th International Workshop on High-pT Physics at LHC
 - “Results from ALICE,” C. Nattrass (ALICE), eConf C1106064. Proceedings for Eleventh Workshop on Non-Perturbative Quantum Chromodynamics 2011.
 - “Di-hadron correlations at RHIC,” C. Nattrass, *Il Nuovo Cimento* 34, 31 (2011). Proceedings, for the Workshop on the Interaction between Soft and Hard 2010.
 - “System size and energy dependence of the near-side of high-p_T triggered di-hadron correlations in STAR,” C. Nattrass (STAR), PoS High-pTphysics09:024 (2009). Proceedings for the 4th International Workshop on High-pT physics at LHC 2009.
 - “Energy and system dependence of high-p_T triggered two-particle near-side correlations,” C. E. Nattrass (STAR), *Eur.Phys.J.C* 62:265-269 (2009). Proceedings for Hot Quarks.
 - “System and energy dependence of strange and non-strange particle correlations in STAR at RHIC,” C. E. Nattrass (STAR), *J. Phys. G* 35 104110 (2008). Proceedings for Quark Matter 2008.
 - “System size dependence of strange particle correlations in Cu+Cu and Au+Au collisions at $\sqrt{s_{NN}} = 200$ GeV in STAR at RHIC,” C. E. Nattrass (STAR), *J. Phys. G* 35 044063 (2008). Proceedings for Strange Quark Matter 2007.

- Invited Talks**
- Southeastern Section of the American Physical Society (Kentucky November 2013)
 - High Pt at the LHC (Grenoble, France November 2013)
 - Southeastern Section of the American Physical Society (Virginia October 2011)
 - Eleventh Workshop on Non-Perturbative Quantum Chromodynamics (Paris, France June 2011)
 - Southeast Conference for Undergraduate Women in Physics (North Carolina January 2011)
 - US LHC User's Meeting (New York October 2010)
 - International Workshop on the Interplay between Soft and Hard Interactions (Catania, Italy September 2010)
 - Southeastern Section of the American Physical Society (Georgia November 2009)
 - 2009 RHIC/AGS Users Meeting ([New York June 2009](#))
 - High-p_T physics at the LHC workshop (Prague, Czech Republic February 2009)
 - BNL workshop on the Ridge (New York September 2008)
 - Hot Quarks (Colorado August 2008)
 - Gordon Conference (New Hampshire June 2008)
 - Gertrude Goldhaber Prize Lecture ([New York May 2008](#))

- Contributed Talks**
- [LBNL Jet Summer School](#) (California June 2010)
 - [Quark Matter](#) (Jaipur, India February 2008)
 - [Trojkráľová Konferencia](#) (Banská Bystrica, Slovakia January 2008)
 - [American Physical Society Division of Nuclear Physics](#) (Virginia October 2007)
 - [Strangeness in Quark Matter](#) (Levoča, Slovakia June 2007)
 - [American Physical Society April Meeting](#) (Florida April 2007)
 - [American Physical Society Division of Nuclear Physics](#) (Tennessee October 2006)
 - American Physical Society March Meeting (Texas March 2003)

- Colloquia**
- Colorado State University (October 2012)
 - University of Mississippi (October 2012)
 - University of Tennessee (February 2012)

- Seminars**
- [Wayne State University](#) (February 2014)
 - Oak Ridge National Laboratory brown bag lunch seminar (March 2012)
 - [Oak Ridge National Laboratory Physics Division](#) (Tennessee June 2011)
 - University of Tennessee at Knoxville (Tennessee April 2011)
 - PHENIX Focus Seminar (New York April 2011)
 - University of Virginia (Virginia March 2011)
 - University of California Davis (California June 2010)

- University of Tennessee at Knoxville (Tennessee November 2009)
 - University of Kentucky (Kentucky October 2009)
 - Lawrence Livermore National Laboratory ([California April 2009](#))
 - University of Tennessee at Knoxville (Tennessee April 2009)
 - Stony Brook University (New York May 2008)
 - Palaver Seminar (Frankfurt, Germany January 2008)
- Poster Presentations**
- Quark Matter (Annecy, France June 2011)
 - RHIC/AGS User's Meeting (New York June 2010)
 - ORNL Women in Science and Engineering Poster Session (Tennessee April 2010)
 - Yale Alumni Conference (Connecticut November 2008)
 - International Conference on Women in Physics ([Seoul, South Korea October 2008](#))
 - Gordon Conference (New Hampshire June 2006)
 - American Physical Society Four Corners meeting (Utah November 2002)
 - American Physical Society Four Corners meeting (New Mexico November 2001)
- Workshop and Conference Organization**
- On organizing committee for the Conference on the Interactions between Particle and Nuclear Physics 2015
 - Co-convenor for “Quark Matter and High Energy Heavy Ion Collisions” session and gave [heavy ion summary talk](#) at the 11th conference on the Intersections of Particle and Nuclear Physics May 28 – June 3, 2012
 - Chair of organizing committee for [Southeast Conference for Undergraduate Women in Physics](#) (January 2012)
 - Gave two day [ALICE Software tutorial](#) directed at undergraduates at Cal Poly (June 2010)
 - Member of organizing committee and graduate panel for the [Southeast Conference for Undergraduate Women in Physics 2010](#)
 - Organizer of [one day meeting](#) for students and post doctoral researchers in heavy ion physics at UTK (November 2009)
 - Member of organizing committees for [Conference for Undergraduate Women in Physics at Yale](#) (January 2008 & January 2009)
 - Member of organizing committee for STAR collaboration meeting (March 2009)
 - Organizer of one day meeting between theorists and experimentalists in heavy ion physics [Yale-Columbia Day at Yale](#) (May 2008)
 - Co-organizer for 2008 RHIC/AGS Users' Meeting workshop on diversity in physics
 - Member of organizing committees for RHIC/AGS User's Meeting (June 2007 & June 2008)
- Service work and training**
- Served as judge for EURECA undergraduate research poster session (Spring 2012)
 - Served as a PHENIX MuID on-call expert (2012 - present)
 - Completed CERN training for working at heights, radiation safety, and working in confined spaces
 - Trained as ALICE shift leader and DAQ/CTP shifter
 - Served as ALICE EMCal on-call expert (2010 - present)
 - Tested and commissioned of front end electronics and LED boards for the ALICE EMCal (2009 - present)
 - Contributed to programming electronics for the PHENIX VTX (2010)
 - Served as computer system administrator for the Yale heavy ion group (2005 - 2009)
 - Contributed to cosmic calibration of ALICE EMCal (2008)
 - Trained as STAR shift leader and detector operator
 - Contributed to the calibration of the STAR Silicon Vertex Tracker (SVT) (2006)
 - Studied V^0 reconstruction with the SVT in the year 5 run for the SVT review (2006)
 - Helped re-cable SVT and make new electronic boards for the read-out system (2006)
 - Studied calibration of SVT using the lasers (2006-2007)
 - Quality assurance for the Year 7 run for the strangeness working group (2008)
- Committee Work**
- Committee on Departmental Undergraduate Advising and Recruiting (Fall 2012 - current)
 - Committee on Undergraduate Curriculum for Majors (Fall 2012 - current)
 - Co-analysis coordinator for transverse energy measurements in ALICE (July 2010 - current)

- Elected Brookhaven National Laboratory (BNL) RHIC/AGS Users' Executive Committee student/post-doctoral fellow representative (June 2006 – June 2008)
- Elected to BNL Association for Students And Post-doctoral fellows board (October 2006 – October 2007)
- Elected Juniors' representative on the STAR council (April 2008 – May 2009)

**Community
Service and
Outreach**

- [Science Cafe talk](#) in Knoxville, TN (April 2013)
- [Science Cafe talk](#) in Oxford, MI (October 2012)
- On organizing committee for panel on The Under-represented Majority at the 2012 of the Southeastern Section of the APS
- Co-organizer of panel on The Under-represented Majority at the 2011 of the Southeastern Section of the APS
- Subject of [ALICE Matters Focus On](#) (February 2011)
- Subject of Knoxville Girl Scout Museum exhibit on former Girl Scouts who are scientists (Fall 2010)
- Served as a blogger for ALICE for [US LHC Blog](#) (April 2010 – January 2011)
- Served on the graduate panel for Southeast Conference for Undergraduate Women in Physics 2010
- Talk for BWIS on [The 3rd International Conference on Women in Physics](#) (April 2009)
- Gave a talk on my career as a woman in physics for the girls at [Riverhead Middle School](#) in panel organized by the Long Island Fund for Women and Girls (April 2009)
- Authored an article in RHIC News January 13, 2009 issue “[After the 3rd International Conference on Women in Physics – What We as a Community Can Do](#)”
- Authored an article in APS Committee on the Status of Women in Physics Gazette Spring 2009 issue “[How the high energy community can help women succeed in physics](#)”
- Served on the graduate panel for the Conference for Undergraduate Women in Physics at Yale 2009
- Participated as a mentor in the Women In Science At Yale program (2008-2009)
- Member of the US delegation to the 2008 IUPAP International Conference on Women in Physics
- Volunteer for Girls' Science Investigations, a science education program for middle school girls (2007-2008)
- Volunteer for BNL Summer Sundays public tours of RHIC and STAR in 2006 and 2007 and PHENIX in 2010

Teaching

- Modern Physics (Fall 2012), Introductory physics for biological scientists I (Spring 2013, Spring 2014) and II (Fall 2013)
- Discussion leader for UTK “Life of the Mind” freshman seminar (2011-2014)
- Undergraduate advisor (Spring 2013 - current)
- Mentor for graduate students at UTK
- Attended Yale graduate school course “Fundamentals of Teaching in the Natural Sciences”
- Level II International Tutor Certification from the College Reading and Learning Association
- Teaching assistant introductory physics lab for biological scientists (Fall 2003, Spring 2004, Fall 2004, Spring 2005, Summer 2005, Fall 2005, Spring 2006, Fall 2006); differential equations (Spring 2007); mathematical methods for physicists (Fall 2007); introductory physics (Spring 2008); Quantum Physics and Beyond seminar (Fall 2008); undergraduate electrodynamics (Spring 2009)
- Supervised Xieyue Fan on her senior thesis completed May 2009
- CSU College of Natural Science Tutorial Hall tutor, Aug. 1999 - Jan. 2000, Aug. 2001 - May 2002, tutoring calculus, differential equations, statistics, and organic chemistry

Research

- Post-doctoral researchers: Natasha Sharma

Advisees

- Graduate students: Joel Mazer, Andy Castro
- Undergraduate: Meg Stuart, Caroline Bowen

Awards

- APS CSWP [Woman Physicist of the Month](#) (May 2012)
- [Gertude Goldhaber Prize](#) to recognize substantial promise and accomplishment by a female graduate student in physics (May 2008)

- American Association of Physics Teachers Teaching Assistant Award (June 2008)
- Certificate of Appreciation from RHIC/AGS Users' Executive Committee (May 2008)
- Barry Goldwater Scholar (2002-2003 school year)